

26020/26021 Chemistry (Polytechnical Foundation) Formula Collection

2026

Contents

1	Atoms, Stoichiometry, and Nomenclature	3
1.1	Atoms and Isotopes	3
1.2	Amounts of Substance	3
1.3	Stoichiometry	3
1.4	Chemical Formulae and Names	4
2	Aqueous Reactions and Gases	4
2.1	Solutions and Electrolytes	4
2.2	Precipitation and Net Ionic Reactions	5
2.3	Redox in Aqueous Solution	5
2.4	Gases	5
3	Thermodynamics	6
3.1	First Law and Enthalpy	6
3.2	Entropy and Gibbs Free Energy	6
4	Electronic Structure and Periodic Table	7
4.1	Light and Quantization	7
4.2	Quantum Numbers and Orbitals	7
4.3	Electron Configurations	8
4.4	Periodic Trends	8
5	Chemical Bonds, Lewis Structures, and VSEPR	9
5.1	Bond Types	9
5.2	Lewis Structures	9
5.3	VSEPR Geometry	9
6	Intermolecular Forces, Solids, and Liquids	10
6.1	Intermolecular Forces	10
6.2	Crystals and Unit Cells	10
6.3	Phase Changes and Vapor Pressure	11
6.4	Colligative Properties	11
7	Chemical Kinetics and Equilibrium	12
7.1	Reaction Rates	12
7.2	Rate Laws	12
7.3	Temperature and Catalysis	12
7.4	Equilibrium	13
8	Organic Chemistry	13
8.1	Hydrocarbons	13

8.2	Isomerism and Nomenclature	14
8.3	Functional Groups	14
8.4	Chirality and Biomolecules	15
9	Acids, Bases, Buffers, and Solubility Products	15
9.1	Bronsted Acids and Bases	15
9.2	Acid and Base Constants	15
9.3	Polyprotic Acids and Buffers	16
9.4	Solubility Products	16
10	Electrochemistry	17
10.1	Oxidation States and Redox	17
10.2	Galvanic Cells	17
10.3	Free Energy and Equilibrium	18
10.4	Nernst Equation	18
11	Exam-Required Extra Topics	18
11.1	Recurring Exam Methods	18

1 Atoms, Stoichiometry, and Nomenclature

1.1 Atoms and Isotopes

1.1.1 Atomic number, mass number, and isotope notation

$${}^A_Z\text{X}, \quad A = Z + N$$

Z is the number of protons, N is the number of neutrons, and A is the mass number. Isotopes have the same Z and different N .

1.1.2 Subatomic particles

particle	relative charge	approximate mass
electron	-1	9.1×10^{-28} g
proton	+1	1.7×10^{-24} g
neutron	0	1.7×10^{-24} g

Atomic charge is determined by proton and electron count. Atomic mass is dominated by protons and neutrons.

1.2 Amounts of Substance

1.2.1 Avogadro constant and mole

$$N = nN_A, \quad N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

The mole converts between a count of particles N and an amount of substance n .

1.2.2 Molar mass relation

$$n = \frac{m}{M}, \quad m = nM$$

m is mass, M is molar mass, and n is amount of substance. This relation is central for converting grams to reaction amounts.

1.2.3 Mass percent

$$w_i = \frac{m_i}{m_{\text{total}}} \times 100\%$$

Mass percent gives the mass fraction of a component in a compound or mixture.

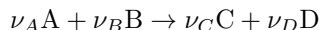
1.2.4 Empirical and molecular formula

$$\text{empirical formula ratio} = \frac{n_i}{\min(n_i)}, \quad \text{molecular formula} = k(\text{empirical formula})$$

The empirical formula gives the smallest integer atom ratio. The molecular formula also requires the molar mass.

1.3 Stoichiometry

1.3.1 Balanced reaction coefficients



Balanced coefficients conserve atoms and charge. Mole ratios in calculations are the ratios of the stoichiometric coefficients.

1.3.2 Limiting reactant

$$\frac{n_i}{\nu_i}$$

The smallest value of n_i/ν_i identifies the limiting reactant. Product amounts are obtained by multiplying this reaction extent by product coefficients.

1.3.3 Theoretical yield

$$n_{\text{product}} = \frac{\nu_{\text{product}}}{\nu_{\text{limiting}}} n_{\text{limiting}}, \quad m_{\text{product}} = n_{\text{product}} M_{\text{product}}$$

The theoretical yield is the maximum product amount allowed by the balanced equation.

1.4 Chemical Formulae and Names

1.4.1 Ionic compound charge neutrality

$$\sum_i q_i n_i = 0$$

The formula of an ionic compound must have total charge zero. Parentheses are used when a polyatomic ion appears more than once, for example $\text{Ca}(\text{HSO}_4)_2$.

1.4.2 Common ion names from lectures and exams

ion	name	ion	name
NH_4^+	ammonium	NO_3^-	nitrate
NO_2^-	nitrite	SO_4^{2-}	sulfate
HSO_4^-	hydrogen sulfate	SO_3^{2-}	sulfite
CO_3^{2-}	carbonate	PO_4^{3-}	phosphate
H_2PO_4^-	dihydrogen phosphate	HPO_4^{2-}	hydrogen phosphate
CN^-	cyanide	OH^-	hydroxide

These ions occur repeatedly in nomenclature, acid–base, redox, and solubility questions.

1.4.3 Molecular compound Greek prefixes

1	mono	2	di	3	tri
4	tetra	5	penta	6	hexa
7	hepta	8	octa	9	nona
10	deca				

For binary molecular compounds, the first element is named first and the second receives the ending “-ide”.

2 Aqueous Reactions and Gases

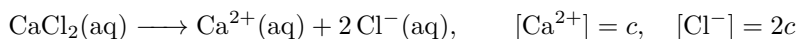
2.1 Solutions and Electrolytes

2.1.1 Molarity

$$c = \frac{n}{V}$$

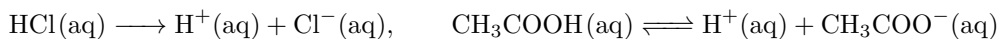
Molarity is amount of substance per solution volume. Formal concentration c is distinguished from actual species concentrations written with brackets.

2.1.2 Actual ion concentrations



Strong electrolytes are treated as fully dissociated in water. Bracket notation gives actual concentration of each dissolved species.

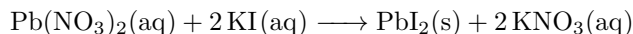
2.1.3 Strong and weak electrolytes



Strong electrolytes ionize essentially completely. Weak electrolytes establish equilibria with both molecular and ionic forms present.

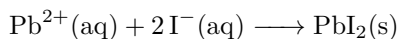
2.2 Precipitation and Net Ionic Reactions

2.2.1 Precipitation reaction



Precipitation occurs when ions form a sparingly soluble solid. Spectator ions can be removed to obtain the net ionic reaction.

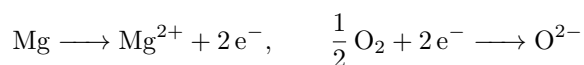
2.2.2 Net ionic equation



The net ionic equation shows only species that chemically change. It is the compact form used for precipitation and acid–base reactions.

2.3 Redox in Aqueous Solution

2.3.1 Redox half-reactions



Oxidation produces electrons and reduction consumes electrons. The electron count must cancel in the summed redox reaction.

2.3.2 Balancing redox reactions

balance atoms except H,O,
balance O with H₂O,
balance H with H⁺ or OH[−],
balance charge with e[−].

In acidic solution use H⁺; in basic solution neutralize added H⁺ with OH[−]. The final equation must conserve atoms and charge.

2.4 Gases

2.4.1 Ideal gas law

$$pV = nRT$$

The ideal gas law relates pressure, volume, amount, and absolute temperature. It is used for gas stoichiometry and pressure changes.

2.4.2 Combined gas law

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

For fixed n , pressure-volume-temperature changes follow this ratio. Temperatures must be in kelvin.

2.4.3 Partial pressure and mole fraction

$$p_i = x_i p_{\text{tot}}, \quad x_i = \frac{n_i}{n_{\text{tot}}}, \quad p_{\text{tot}} = \sum_i p_i$$

Dalton's law expresses total pressure as the sum of component partial pressures.

2.4.4 Gas density and molar mass

$$\rho = \frac{m}{V} = \frac{pM}{RT}, \quad M = \frac{\rho RT}{p}$$

Gas density can determine molar mass when the gas behaves ideally.

2.4.5 Graham's law of diffusion

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

At equal temperature, lighter gases diffuse or effuse faster than heavier gases.

3 Thermodynamics

3.1 First Law and Enthalpy

3.1.1 First law of thermodynamics

$$\Delta U = q + w$$

Internal energy changes by heat q and work w . Heat absorbed by the system is positive; heat released is negative.

3.1.2 Pressure-volume work

$$w = -p_{\text{ext}}\Delta V$$

Expansion work is negative for the system when $\Delta V > 0$. The lecture sign convention connects constant-pressure heat to enthalpy.

3.1.3 Enthalpy

$$H = U + pV, \quad q_p = \Delta H$$

At constant pressure, the heat exchanged equals the enthalpy change. Enthalpy is a state function.

3.1.4 Reaction enthalpy

$$\Delta H_{\text{rxn}} = H_{\text{products}} - H_{\text{reactants}}$$

$\Delta H_{\text{rxn}} < 0$ is exothermic and $\Delta H_{\text{rxn}} > 0$ is endothermic.

3.1.5 Standard enthalpy of formation

$$\Delta H_{\text{rxn}}^{\circ} = \sum_i \nu_i \Delta H_{\text{f}}^{\circ}(\text{products}) - \sum_i \nu_i \Delta H_{\text{f}}^{\circ}(\text{reactants})$$

Elements in their most stable standard state have $\Delta H_{\text{f}}^{\circ} = 0$. Coefficients from the balanced reaction multiply formation enthalpies.

3.1.6 Hess's law

$$\Delta H_{\text{overall}}^{\circ} = \sum_k \Delta H_k^{\circ}$$

Enthalpy changes add because enthalpy is a state function. Reversing a reaction changes the sign of ΔH .

3.2 Entropy and Gibbs Free Energy

3.2.1 Gibbs free energy

$$\Delta G = \Delta H - T\Delta S$$

Gibbs free energy combines enthalpy and entropy at temperature T . Use T in kelvin and consistent energy units.

3.2.2 Spontaneity criterion

$$\Delta G < 0 : \text{spontaneous}, \quad \Delta G = 0 : \text{equilibrium}, \quad \Delta G > 0 : \text{nonspontaneous}$$

The sign of ΔG predicts thermodynamic direction under the specified conditions.

3.2.3 Temperature dependence of spontaneity

$$\Delta G(T) = \Delta H - T\Delta S$$

If ΔH and ΔS are treated as constant, increasing T changes ΔG through the entropy term.

3.2.4 Standard reaction entropy

$$\Delta S_{\text{rxn}}^\circ = \sum_i \nu_i S^\circ(\text{products}) - \sum_i \nu_i S^\circ(\text{reactants})$$

Entropy changes are calculated from standard molar entropies using the balanced reaction.

3.2.5 Thermochemical equation scaling



Thermodynamic quantities for a reaction scale with the written chemical equation.

4 Electronic Structure and Periodic Table

4.1 Light and Quantization

4.1.1 Photon energy

$$E = h\nu = \frac{hc}{\lambda}$$

Photon energy is proportional to frequency and inversely proportional to wavelength. h is Planck's constant.

4.1.2 Photoelectric effect

$$h\nu = E_{\text{kin}} + W$$

W is the work function of the metal. Electrons are emitted only when the photon energy exceeds the threshold.

4.1.3 Bohr energy levels

$$E_n = -\frac{R_{\text{H}}}{n^2}, \quad n = 1, 2, 3, \dots$$

Allowed electron energies in hydrogen are discrete. Transitions absorb or emit energy differences between levels.

4.1.4 De Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Matter with momentum p has wave character. This motivates standing-wave orbitals for electrons.

4.2 Quantum Numbers and Orbitals

4.2.1 Quantum numbers

symbol	allowed values	meaning
n	$1, 2, 3, \dots$	shell
ℓ	$0, 1, \dots, n-1$	subshell shape
m_ℓ	$-\ell, \dots, +\ell$	orbital orientation
m_s	$\pm \frac{1}{2}$	spin

One electron in an atom is described by a unique set of four quantum numbers.

4.2.2 Subshell labels

$$\ell = 0, 1, 2, 3 \quad \leftrightarrow \quad s, p, d, f$$

For a subshell with angular momentum ℓ , there are $2\ell + 1$ orbitals.

4.2.3 Orbital capacity

$$\text{maximum electrons in a subshell} = 2(2\ell + 1)$$

Each orbital holds at most two electrons with opposite spin.

4.3 Electron Configurations

4.3.1 Aufbau ordering

$$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, \dots$$

Electrons occupy lower-energy orbitals first. The $n + \ell$ rule gives the approximate filling order.

4.3.2 Pauli exclusion principle

$$\text{no two electrons in an atom have the same } (n, \ell, m_\ell, m_s)$$

The Pauli principle limits each orbital to two electrons.

4.3.3 Hund's rule

$$\text{within degenerate orbitals, maximize unpaired parallel spins}$$

Hund's rule gives the lowest-energy arrangement before pairing occurs.

4.4 Periodic Trends

4.4.1 Effective nuclear charge

$$Z_{\text{eff}} \approx Z - S$$

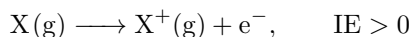
Z_{eff} is the nuclear attraction felt after shielding by other electrons. Across a period, Z_{eff} generally increases.

4.4.2 Atomic radius trend

$$\text{across a period: radius decreases,} \quad \text{down a group: radius increases}$$

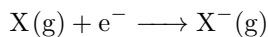
Increasing effective nuclear charge contracts atoms across a period. Additional shells increase radius down a group.

4.4.3 Ionization energy



Ionization energy is the endothermic energy required to remove an electron from a gaseous atom.

4.4.4 Electron affinity



Electron affinity describes the energy change when a gaseous atom accepts an electron.

4.4.5 Mulliken electronegativity

$$\chi = \frac{\text{IE} + \text{EA}}{2}$$

Electronegativity measures tendency to attract electron density in a bond. Fluorine is the most electronegative element in the lecture material.

4.4.6 Isoelectronic ionic radius

$$\text{same electron count: larger } Z \Rightarrow \text{smaller radius}$$

For isoelectronic species, increasing nuclear charge pulls the same electron cloud closer.

5 Chemical Bonds, Lewis Structures, and VSEPR

5.1 Bond Types

5.1.1 Ionic bonding and Coulomb stabilization

$$E \propto \frac{q_1 q_2}{r}$$

Ionic solids are stabilized by electrostatic attraction between oppositely charged ions. Lattice enthalpy can make salt formation strongly favorable.

5.1.2 Born-Haber cycle

$$\Delta H_f^\circ = \sum \Delta H_{\text{atomization/ionization/electron affinity}}^\circ + \Delta H_{\text{lattice}}^\circ$$

A Born-Haber cycle applies Hess's law to ionic solid formation. Large negative lattice enthalpy stabilizes the crystal.

5.1.3 Covalent bond polarity

$$\Delta\chi = |\chi_A - \chi_B|$$

Greater electronegativity difference gives a more polar bond. Molecular polarity also depends on geometry.

5.1.4 Molecular orbital bond order

$$\text{bond order} = \frac{N_{\text{bonding}} - N_{\text{antibonding}}}{2}$$

Bond order estimates bond strength and whether a simple diatomic molecule is stabilized.

5.2 Lewis Structures

5.2.1 Valence electron count

$$N_{\text{val}} = \sum_i N_{\text{val},i} - q$$

For an ion, subtract positive charge and add negative charge. The total electron count controls the Lewis structure.

5.2.2 Formal charge

$$\text{FC} = V - \left(N_{\text{nonbonding}} + \frac{1}{2} N_{\text{bonding}} \right)$$

Formal charge compares assigned electrons to neutral valence electrons. Favored resonance forms usually minimize charge and place negative charge on electronegative atoms.

5.2.3 Octet rule and expanded octet

second-period atoms normally obey an octet

Atoms from period 3 and below may appear with expanded octets in course examples such as SF_6 . Electron-deficient molecules such as BH_3 are exceptions.

5.2.4 Resonance structures

real structure = resonance hybrid of valid Lewis forms

Resonance forms differ only in electron placement, not atom positions. Equivalent resonance forms imply delocalized bonding and equalized bond lengths.

5.3 VSEPR Geometry

5.3.1 Electron-density domains

steric number = number of bonded atoms + number of lone pairs

VSEPR arranges electron domains to maximize separation around the central atom.

5.3.2 Common VSEPR geometries

steric number	lone pairs	molecular geometry	example
2	0	linear	CO ₂
3	0	trigonal planar	BH ₃
4	0	tetrahedral	CH ₄
4	1	trigonal pyramidal	NH ₃
4	2	bent	H ₂ O
5	0	trigonal bipyramidal	PCl ₅
6	0	octahedral	SF ₆
6	1	square pyramidal	IF ₅
5	3	linear	XeF ₂

The exam wording often asks for planar, bent, pyramidal, tetrahedral, or octahedral geometry.

5.3.3 Representative bond angles

geometry	ideal angle
linear	180°
trigonal planar	120°
tetrahedral	109.5°
trigonal pyramidal	≈ 107°
bent water-like	≈ 105°
octahedral	90°

Lone pairs compress bonding angles relative to ideal electron-domain angles.

5.3.4 Molecular polarity from bond dipoles

$$\vec{\mu}_{\text{mol}} = \sum_i \vec{\mu}_i$$

Polar bonds can cancel in symmetric molecules. Molecular polarity depends on both electronegativity and molecular shape.

6 Intermolecular Forces, Solids, and Liquids

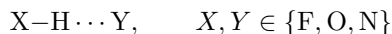
6.1 Intermolecular Forces

6.1.1 Van der Waals interactions

dispersion and dipole interactions: typically $< 10 \text{ kJ mol}^{-1}$

Van der Waals forces act between permanent or induced dipoles. They dominate noble gas liquids and nonpolar molecular substances.

6.1.2 Hydrogen bonding



Hydrogen bonding requires hydrogen bound to F, O, or N interacting with F, O, or N. It raises boiling points and stabilizes structures such as ice and biomolecules.

6.1.3 Like dissolves like

polar dissolves polar, nonpolar dissolves nonpolar

Solubility is favored when solute-solvent interactions can replace solute-solute and solvent-solvent interactions.

6.2 Crystals and Unit Cells

6.2.1 Crystal lattice and unit cell

crystal = periodic translations of a unit cell

A unit cell is the smallest repeating volume used to describe a crystalline lattice.

6.2.2 Atoms per cubic unit cell

cell type	atoms per cell	packing fraction
simple cubic	1	0.52
body-centered cubic	2	0.68
face-centered cubic	4	0.74

Corner atoms contribute 1/8, face atoms 1/2, and body-center atoms 1 to one unit cell.

6.2.3 Density of a unit cell

$$\rho = \frac{ZM}{N_A a^3}$$

Z is atoms or formula units per unit cell and a is the cubic cell edge length. Use consistent units for a^3 .

6.2.4 Cubic cell radius relations

cell type	radius relation
simple cubic	$a = 2r$
body-centered cubic	$\sqrt{3}a = 4r$
face-centered cubic	$\sqrt{2}a = 4r$

These relations connect atomic radius and cell edge length for packed spheres.

6.3 Phase Changes and Vapor Pressure

6.3.1 Phase transition enthalpies

$$\Delta H_{\text{sub}} = \Delta H_{\text{fus}} + \Delta H_{\text{vap}}$$

Sublimation can be viewed as fusion followed by vaporization. Fusion and vaporization enthalpies are positive for the forward phase changes.

6.3.2 Clausius-Clapeyron equation

$$\ln P = -\frac{\Delta H_{\text{vap}}}{RT} + C$$

Vapor pressure increases with temperature. A plot of $\ln P$ versus $1/T$ has slope $-\Delta H_{\text{vap}}/R$.

6.3.3 Two-point Clausius-Clapeyron equation

$$\ln \left(\frac{P_2}{P_1} \right) = -\frac{\Delta H_{\text{vap}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

This form eliminates the constant C and is used to calculate vapor pressure or boiling point.

6.4 Colligative Properties

6.4.1 Raoult's law

$$P_1 = X_1 P_1^\circ = (1 - X_2) P_1^\circ$$

A nonvolatile solute lowers the solvent vapor pressure. X_1 is the solvent mole fraction.

6.4.2 Molality

$$m = \frac{n_{\text{solute}}}{m_{\text{solvent}}(\text{kg})}$$

Molality is independent of temperature because it uses solvent mass, not solution volume.

6.4.3 Freezing point depression and boiling point elevation

$$\Delta T_f = iK_f m, \quad \Delta T_b = iK_b m$$

K_f and K_b are solvent constants. The van't Hoff factor i accounts for dissolved particles when dissociation is relevant.

7 Chemical Kinetics and Equilibrium

7.1 Reaction Rates

7.1.1 Stoichiometric reaction rate

$$aA + bB \rightarrow cC + dD$$
$$r = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

The reaction rate is normalized by stoichiometric coefficients so all species describe the same reaction progress.

7.1.2 Average and instantaneous rate

$$r_{\text{avg}} \approx -\frac{\Delta[A]}{\Delta t}, \quad r = -\frac{d[A]}{dt}$$

The instantaneous rate is the slope of the concentration-time curve at a point.

7.2 Rate Laws

7.2.1 General rate law

$$r = k[A]^m[B]^n$$

The exponents are experimentally determined reaction orders. The overall order is $m + n$.

7.2.2 First-order integrated rate law

$$[A]_t = [A]_0 e^{-kt}, \quad \ln[A]_t = -kt + \ln[A]_0$$

A first-order plot of $\ln[A]$ versus t is linear with slope $-k$.

7.2.3 Second-order integrated rate law

$$\frac{1}{[A]_t} = kt + \frac{1}{[A]_0}$$

A second-order plot of $1/[A]$ versus t is linear with slope k .

7.2.4 Half-life

$$t_{1/2}^{(1)} = \frac{\ln 2}{k}, \quad t_{1/2}^{(2)} = \frac{1}{k[A]_0}$$

First-order half-life is independent of initial concentration. Second-order half-life depends on $[A]_0$.

7.3 Temperature and Catalysis

7.3.1 Arrhenius equation

$$k = Ae^{-E_a/RT}$$

E_a is activation energy and A is the frequency factor. Higher temperature increases the fraction of molecules above the activation barrier.

7.3.2 Linear Arrhenius form

$$\ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A$$

A plot of $\ln k$ versus $1/T$ has slope $-E_a/R$.

7.3.3 Two-temperature Arrhenius relation

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_a}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

This relation compares rate constants at two temperatures when E_a is constant.

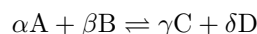
7.3.4 Catalyst effect

$$E_{a,cat} < E_{a,uncat}$$

A catalyst lowers the activation barrier and increases rate. It does not change the thermodynamic equilibrium constant.

7.4 Equilibrium

7.4.1 Reaction quotient



$$Q = \frac{a_C^\gamma a_D^\delta}{a_A^\alpha a_B^\beta}$$

Activities are usually represented by concentrations for solutes and partial pressures for gases in this course. Pure solids, pure liquids, and solvents are omitted.

7.4.2 Equilibrium constant

$$K = Q_{eq}$$

At equilibrium, forward and reverse rates balance and the reaction quotient equals K .

7.4.3 Gibbs free energy and reaction quotient

$$\Delta G = \Delta G^\circ + RT \ln Q$$

Non-standard spontaneity is determined by ΔG , not by ΔG° alone.

7.4.4 Standard free energy and equilibrium constant

$$\Delta G^\circ = -RT \ln K, \quad K = e^{-\Delta G^\circ / RT}$$

$K > 1$ corresponds to $\Delta G^\circ < 0$ and product-favored equilibrium.

7.4.5 Le Chatelier's principle

disturbance $\Rightarrow$ shift opposing the disturbance

Changes in concentration, pressure, volume, or temperature shift an equilibrium to reduce the imposed change.

7.4.6 Gas equilibrium constant

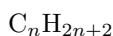
$$K_p = \frac{(p_C)^\gamma (p_D)^\delta}{(p_A)^\alpha (p_B)^\beta}$$

For gas equilibria, partial pressures are used in the same stoichiometric powers as concentration expressions.

8 Organic Chemistry

8.1 Hydrocarbons

8.1.1 Alkanes



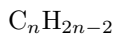
Alkanes are saturated hydrocarbons with only single bonds. They are named by the longest carbon chain and alkyl substituents.

8.1.2 Alkenes



Alkenes contain a carbon-carbon double bond. Restricted rotation can give cis/trans isomerism when each alkene carbon has different substituents.

8.1.3 Alkynes



Alkynes contain a carbon-carbon triple bond. C_2H_2 is the simplest alkyne Lewis-structure example in the slides.

8.1.4 Aromatic hydrocarbons



Benzene is a planar six-membered aromatic ring with one hydrogen on each carbon. Aromatic rings are recurring structural motifs in functional-group questions.

8.2 Isomerism and Nomenclature

8.2.1 Structural isomers

same molecular formula, different atom connectivity

Isomers have different structures and therefore different properties. Alkane isomer counting appears in lecture exercises.

8.2.2 Alkane naming

locants + substituents + parent chain

Choose the longest carbon chain, number from the end nearest substituents, and list substituents alphabetically.

8.2.3 Degree of unsaturation

$$DBE = \frac{2C + 2 + N - H - X}{2}$$

The double-bond equivalent counts rings and π -bonds for formulas containing C, H, N, and halogens X . Oxygen does not affect the count.

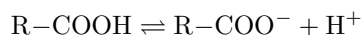
8.3 Functional Groups

8.3.1 Functional group recognition

group	structural motif
alcohol	$R-OH$
ether	$R-O-R'$
aldehyde	$R-CHO$
ketone	$R-CO-R'$
carboxylic acid	$R-COOH$
ester	$R-COO-R'$
amine	$R-NH_2, R_2NH, R_3N$
amide	$R-CONH_2, R-CONHR', R-CONR'_2$

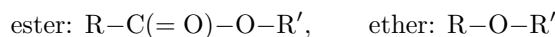
Functional group identification is a recurring exam concept, especially in molecules with several groups.

8.3.2 Carboxylic acid deprotonation



Carboxylic acids contain a COOH group and form carboxylate conjugate bases.

8.3.3 Ester and ether distinction



An ester contains a carbonyl next to oxygen. An ether contains an oxygen single-bonded to two carbon groups.

8.4 Chirality and Biomolecules

8.4.1 Chiral center

C^* = tetrahedral carbon attached to four different groups

Interchanging two groups at a chiral center gives the enantiomer. Chirality was highlighted with ibuprofen and amino-acid structures.

8.4.2 Amino acids and peptides



Amino acids contain amine and carboxylic acid functional groups. Peptides form by connecting amino acids through amide bonds.

8.4.3 Degree of polymerization

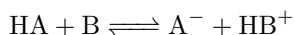
$$n = \frac{M_{\text{polymer}}}{M_{\text{repeat}}}$$

The degree of polymerization is the number of repeat units in a polymer chain.

9 Acids, Bases, Buffers, and Solubility Products

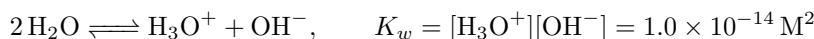
9.1 Bronsted Acids and Bases

9.1.1 Bronsted definition



A Bronsted acid donates a proton and a Bronsted base accepts a proton. Products are the conjugate base and conjugate acid.

9.1.2 Autodissociation of water



At 298 K, neutral water has $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$.

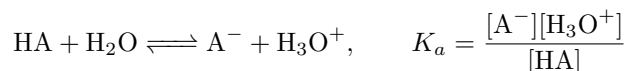
9.1.3 pH and pOH

$$\text{pH} = -\log[\text{H}_3\text{O}^+], \quad \text{pOH} = -\log[\text{OH}^-], \quad \text{pH} + \text{pOH} = 14$$

The pH scale is logarithmic. A decrease of one pH unit means ten times larger $[\text{H}_3\text{O}^+]$.

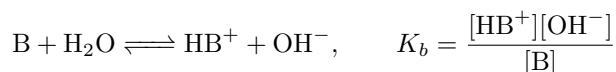
9.2 Acid and Base Constants

9.2.1 Acid ionization constant



Larger K_a means stronger acid. $\text{p}K_a = -\log K_a$.

9.2.2 Base ionization constant



Larger K_b means stronger base. $\text{p}K_b = -\log K_b$.

9.2.3 Conjugate acid-base relation

$$K_a(\text{HA})K_b(\text{A}^-) = K_w, \quad pK_a + pK_b = pK_w = 14$$

Conjugate acid-base strength is linked through water autoionization at 298 K.

9.2.4 Weak acid approximation

$$[\text{H}_3\text{O}^+] \approx \sqrt{K_a c_0}, \quad \text{pH} \approx -\log \sqrt{K_a c_0}$$

This approximation applies when dissociation is small compared with the formal acid concentration c_0 .

9.2.5 Weak base approximation

$$[\text{OH}^-] \approx \sqrt{K_b c_0}, \quad \text{pOH} \approx -\log \sqrt{K_b c_0}$$

This approximation applies when base protonation is small compared with the formal base concentration c_0 .

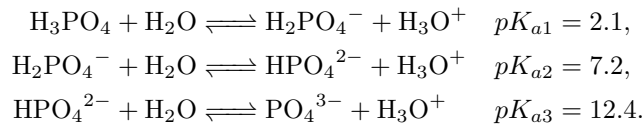
9.2.6 Strong acid and strong base pH

$$\text{strong acid: } [\text{H}_3\text{O}^+] \approx c_{\text{acid}}, \quad \text{strong base: } [\text{OH}^-] \approx \nu_{\text{OH}^-} c_{\text{base}}$$

Strong electrolytes are treated as completely dissociated. Stoichiometry matters for bases such as $\text{Ba}(\text{OH})_2$.

9.3 Polyprotic Acids and Buffers

9.3.1 Phosphoric acid equilibria



The relevant conjugate pair is selected by the pH region.

9.3.2 Henderson-Hasselbalch equation

$$\text{pH} = pK_a + \log \left(\frac{[\text{A}^-]}{[\text{HA}]} \right)$$

Buffers contain a weak acid and its conjugate base. The pH is closest to pK_a when the two concentrations are comparable.

9.3.3 Buffer after acid or base addition



Small additions are handled first by stoichiometric neutralization, then by Henderson-Hasselbalch with the new amounts.

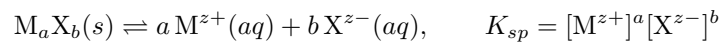
9.3.4 Titration equivalence point

$$n_{\text{acid equivalents}} = n_{\text{base equivalents}}$$

At equivalence, stoichiometric acid-base equivalents are equal. The pH depends on whether the remaining salt is acidic, basic, or neutral.

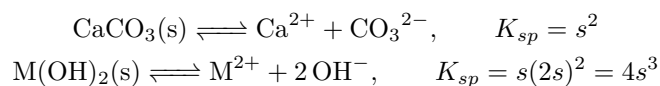
9.4 Solubility Products

9.4.1 Solubility product



Pure solid is omitted from the equilibrium expression. K_{sp} has concentration powers set by stoichiometry.

9.4.2 Molar solubility from K_{sp}



The molar solubility s is converted to ion concentrations using the dissolution stoichiometry.

9.4.3 Ion product and precipitation

$$Q_{sp} = [\text{M}^{z+}]^a [\text{X}^{z-}]^b$$

If $Q_{sp} > K_{sp}$, precipitation is favored. If $Q_{sp} < K_{sp}$, more solid can dissolve.

9.4.4 Common-ion effect on solubility

$$K_{sp} = [\text{M}^{z+}][\text{X}^{z-}]$$

Adding an ion already present in the dissolution equilibrium lowers the molar solubility by shifting the equilibrium toward the solid.

10 Electrochemistry

10.1 Oxidation States and Redox

10.1.1 Oxidation state rules

free element	0
monoatomic ion	ion charge
oxygen in most compounds	$-II$
hydrogen in most compounds	$+I$
halogens in binary salts	$-I$
sum in molecule or ion	overall charge

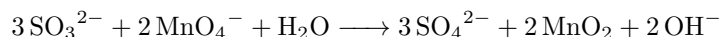
Oxidation states are bookkeeping numbers used to identify oxidation and reduction.

10.1.2 Oxidation and reduction

oxidation: loss of electrons, reduction: gain of electrons

Oxidation state increases during oxidation and decreases during reduction.

10.1.3 Basic-solution redox balancing



This lecture example conserves atoms and charge in basic solution. It is a reusable check for half-reaction balancing.

10.2 Galvanic Cells

10.2.1 Anode and cathode

anode: oxidation, cathode: reduction

In a galvanic cell, electrons flow from anode to cathode through the external circuit.

10.2.2 Cell diagram notation



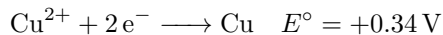
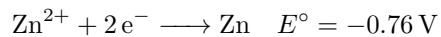
A single vertical line denotes a phase boundary and the double line denotes a salt bridge.

10.2.3 Standard cell potential

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ} = E_{\text{red}}^{\circ}(\text{reduction}) - E_{\text{red}}^{\circ}(\text{oxidation})$$

Standard reduction potentials are used as tabulated. The more positive reduction potential is reduced in a spontaneous galvanic cell.

10.2.4 Daniell cell potential



$$E_{\text{cell}}^{\circ} = 0.34 - (-0.76) = 1.10\text{ V}$$

Zinc is oxidized and copper(II) is reduced in the Daniell cell.

10.3 Free Energy and Equilibrium

10.3.1 Electrical work and Gibbs free energy

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

n is moles of electrons transferred and $F = 9.647 \times 10^4 \text{ C mol}^{-1}$. Positive E_{cell}° gives negative ΔG° .

10.3.2 Cell potential and equilibrium constant

$$nFE_{\text{cell}}^{\circ} = RT \ln K$$

At 298 K, a larger standard cell potential corresponds to a larger equilibrium constant.

10.4 Nernst Equation

10.4.1 Nernst equation

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{nF} \ln Q$$

The cell potential changes with reaction quotient Q . Products in Q lower the potential when they increase.

10.4.2 Nernst equation at 298 K

$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.0592\text{ V}}{n} \log Q$$

This base-10 form is used for room-temperature calculations.

10.4.3 Concentration cell

$$E_{\text{cell}} = \frac{0.0592\text{ V}}{n} \log \left(\frac{c_{\text{high}}}{c_{\text{low}}} \right)$$

Identical electrodes give $E_{\text{cell}}^{\circ} = 0$. The spontaneous direction reduces ions in the more concentrated half-cell and dissolves metal in the dilute half-cell.

10.4.4 Solubility product from cell potential

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ} = -RT \ln K$$

Electrochemical measurements can determine equilibrium constants, including K_{sp} , when the cell reaction is written for the dissolution or precipitation equilibrium.

11 Exam-Required Extra Topics

11.1 Recurring Exam Methods

11.1.1 Ideal gas stoichiometry

$$n_{\text{gas}} = \frac{pV}{RT}$$

Gas volumes in exam stoichiometry are converted to moles before applying balanced reaction coefficients.

11.1.2 Combustion enthalpy from formation enthalpies

$$\Delta H_{\text{comb}}^{\circ} = \sum \nu \Delta H_{\text{f}}^{\circ}(\text{CO}_2, \text{H}_2\text{O}, \dots) - \sum \nu \Delta H_{\text{f}}^{\circ}(\text{fuel}, \text{O}_2)$$

The standard formation enthalpy of $\text{O}_2(\text{g})$ is zero. Combustion questions require a balanced combustion reaction first.

11.1.3 Percent change in hydronium concentration from pH

$$\frac{[\text{H}_3\text{O}^+]_2}{[\text{H}_3\text{O}^+]_1} = 10^{\text{pH}_1 - \text{pH}_2}, \quad \% \text{ increase} = (\text{ratio} - 1)100\%$$

This method follows directly from the logarithmic pH definition and appears in ocean-acidification style questions.